College of Engineering – Department of Electronics Engineering

CURRICULUM

Bachelor of Science in Biomedical Engineering

Academic Year 2025-2026

Reference CMOs: CMO No. 86, s. 2017, CMO No. 20 s. 2013, CMO No. 39, s.2021, and CMO No. 40 s. 2021

Curriculum Description

Biomedical Engineering is an interdisciplinary field that integrates biology and engineering to provide technical innovations in medical devices, procedures, and patient care. It aims to bridge the gap between engineering principles and biological and medical sciences, leading to significant advancements in healthcare.

The central philosophy of this curriculum is integration—integrating foundational engineering knowledge with biological sciences, theoretical understanding with hands-on practice, and academic training with the expectations and challenges of the biomedical industry. By embedding industrial exposure, internships, project-based learning, and collaborations with medical and research institutions into the academic framework, students are better equipped to transition seamlessly into the workforce or advanced research roles.

Biomedical Engineering promotes innovation in healthcare by developing new medical devices, technologies, and therapies that improve diagnosis, treatment, and patient care, leading to more effective healthcare solutions.

Beyond healthcare, Biomedical Engineering also plays a critical role in manufacturing, which creates advanced materials; sports medicine, which develops innovative sports equipment and improves athletic recovery and performance; and agriculture.

Program Education Objectives

The chemical engineering alumni three to five years after graduation shall:

PEO 1: Practiced as specialist in solving complex biomedical engineering problems leading to improvements and innovations, while taking into consideration the environmental, social, and economical requirements.
PEO 2: Assumed leadership position in industry, academe, government, or private sector within consideration to social and ethical responsibility.
PEO 3: Engaged in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:
SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex biomedical engineering problems;

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SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;

SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.

SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SO5 Problem Analysis. Identify, formulate, and solve complex biomedical engineering problems by applying principles of engineering, science, and mathematics;

SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex biomedical engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex biomedical engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Probability and Statistics for AI	3	0	3
Differential Equations	3	0	3
Advanced Engineering Mathematics for BioE	3	0	3
Subtotal	**17**	**0**	**17**
B. Natural/ Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers	3	3	4
Organic and Biochemistry	3	3	4
Modern Biology	3	0	3

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Subtotal	**12**	**9**	**15**
C. Basic Engineering Sciences			
Introduction to Engineering	0	3	1
Computer Programming	0	3	1
Engineering Drawing	0	3	1
Computer-Aided Design	0	3	1
Basic Occupational Safety and Health	3	0	3
Technopreneurship	3	0	3
Professional Orientation	2	0	2
Environmental Science and Engineering	3	0	3
Subtotal	**11**	**12**	**15**
D. Allied Courses			
Basic Electrical Engineering	2	3	3
Electronic Circuits: Devices and Analysis	3	3	4
Microprocessor and Microcontroller Systems and Design	3	3	4
Subtotal	**8**	**9**	**11**
E. Professional Courses			
Principles of Molecular Biology	3	0	3
Human Physiology	3	0	3
Computational Bioengineering	2	3	3
Mechanics of Biological System	3	0	3
Introduction to Physiological Control System	3	0	3
Biofluid Mechanics	3	0	3
Bioinstrumentation and Devices	2	3	3
Muskoskeletal Mechanics	3	0	3
BioE Capstone Design 1	0	3	1
BioE Capstone Design 2	0	3	1
Biomedical Laws and Professional Ethics	3	0	3
Biomedical Electronics	3	3	4
Programming and Modelling for BioE	0	3	1
Biomedical Signal Processing	3	3	4
Robotics for Biomedical Applications	3	3	4
Supply Chain Management for Biomedical Engineering	0	0	3
BioE Elective 1	2	3	3
BioE Elective 2	2	3	3
BioE Elective 3	2	3	3
Subtotal	**43**	**33**	**54**
F. Industry Placement			
Industry Placement 1	640 Hours		**8**
Industry Placement 2	640 Hours		**8**
Subtotal			**16**
TOTAL TECHNICAL COURSES	**91**	**63**	**128**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
Understanding the Self	3	0	3
Mathematics in the Modern World	3	0	3
The Contemporary World	3	0	3

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Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology and Society	3	0	3
Subtotal	**24**	**0**	**24**
B. General Education Electives			
Life and Works of Rizal	3	0	3
Linear Algebra for AI	3	0	3
Machine Learning and Neural Network	2	3	3
Entrepreneurial Mind	3	0	3
Subtotal	**11**	**3**	**12**
C. Physical Activities Toward Health and Fitness			
PATHFit 1 - Movement Competency Training	2	0	2
PATHFit 2 - Exercise-based Fitness Activities	2	0	2
PATHFit 3 – Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities 1	2	0	2
PATHFit 4 – Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities 2)	2	0	2
Subtotal	**8**	**0**	**8**
D. National Service Training Program			
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Sub-total	**6**	**0**	**6**
E. Institutional Mandated Course			
Foreign Language	0	3	3
Sub Total	0	**3**	**3**
TOTAL NON-TECHNICAL COURSES	**52**	**3**	**53**
GRAND TOTAL	**143**	**66**	**181**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	17
B. Natural/Physical Sciences	15
C. Basic Engineering Sciences	15
D. Allied Courses	11
E. Professional Courses	54
F. Industry Placements	16
II. NON-TECHNICAL COURSES	
A. General Education Courses	24
B. Filipino/Literature/Mandated Courses	12
C. PATHFit 1-4	8
D. NSTP	6
E. Institutional Mandated Course	3
GRAND TOTAL	**181**

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CURRICULUM STRUCTURE

FIRST YEAR						
First Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
GEd 101	Understanding the Self	3	3	0		
GEd 102	Mathematics in the Modern World	3	3	0		
GEd 105	Readings in Philippine History	3	3	0		
GEd 106	Purposive Communication	3	3	0		
CHEM 101	Chemistry for Engineers	4	3	3		
MATH 101	Differential Calculus	4	4	0		
ENGG 101	**Introduction to Engineering*** **	**1**	**0**	**3**		
PATHFit 1	Movement Competency Training	2	2	0		
NSTP 111	National Service Training Program 1	3	3	0		
	Total	26	24	6		

FIRST YEAR						
Second Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
GEd 104	The Contemporary World	3	3	0		
GEd 108	Art Appreciation	3	3	0		
GEd 109	**Science, Technology and Society*** **	**3**	**3**	**0**		
CpE 101	Computer Programming	1	0	3		
MATH 102	Integral Calculus	4	4	0	MATH 101	
ENGG 102	Engineering Drawing	1	0	3		

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PHYS 101	Physics for Engineers	4	3	3	MATH 101	MATH 102
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1	
NSTP 121	National Service Training Program 2	3	3	0	NSTP lll	
	Total	24	21	9		

FIRST YEAR						
Midterm						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
MATH 103	Differential Equations	3	3	0	MATH 102	
BioE 101	Modern Biology	3	3	0		
IMC 101	Foreign Language	3	3	0		
	Total	9	9	0		
SECOND YEAR						
First Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
BioE 201	Organic and Biochemistry	4	3	3	BioE 101	
BioE 202	Human Physiology	3	3	0	BioE 101	
AI 101	Linear Algebra for AI	3	3	0		
GEd 107	Ethics	3	3	0		
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102	
EE 205	Basic Electrical Engineering	3	2	3	PHYS 101	
ECE 405	Electronic Circuits: Devices and Analysis	4	3	3		EE 205
ENGG 113	**Environmental Science and Engineering *****	**3**	**3**	**0**		

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PATHFit 3	PATHFit 3 – Menu of Dance, Sports, Martial Arts, Group Exercise, Outdoor Adventure Activities Menu Offering: (Traditional and Recreational Games)	2	2	0	PATHFit 1, PATHFit 2	
	Total	26	22	12		
SECOND YEAR						
Second Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
BioE 203	Principles of Molecular Biology	3	3	0	BioE 201	
BioE 204	Advanced Engineering Mathematics for BioE	3	3	0	MATH 103	
BioE 205	Computational Bioengineering	3	2	3	CpE 101	BioE 204
BioE 206	**Biomedical Electronics *****	**4**	**3**	**3**	**ECE 405**	
BioEE 401	BioE Elective 1	3	2	3		
ENGG 414	Basic Occupational Safety and Health	3	3	0		
PATHFit 4	PATHFit 4 – Menu of Dance, Sports, Martial Arts, Group Exercise, Outdoor Adventure Activities Menu Offering: (Team Sports, Basketball and Volleyball)	2	2	0	PATHFit 3	
AI 102	Probability and Statistics for AI	3	3	0	MATH 103	
ENGG 125	Professional Orientation	2	2	0		
	Total	26	23	9		
THIRD YEAR						
First Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
ENGG 126	Industry Placement 1	8	640 hours		3rd year standing	
	Total	8	0	0		
THIRD YEAR						
Second Semester						

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Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
AI 103	Machine Learning and Neural Networks	3	2	3	AI 102	
BioE 308	Supply Chain Management for Biomedical Engineering	3	3	0	MATH 102	
BioE 301	Introduction to Physiological Control System	3	3	0	BioE 202, BioE 204	
BioE 302	Mechanics of Biological Systems	3	3	0	BioE 202, ENGG 409	
BioE 303	Biofluid Mechanics	3	3	0	PHYS 101	BioE 302
BioE 304	**Bioinstrumentation and Devices *****	**3**	**2**	**3**	**BioE 206**	
BioE 305	Biomedical Signal Processing	4	3	3	BioE 204	
BioE 403	BioE Capstone Design 1	1	0	3	4th Year Standing	
BioEE 402	BioE Elective 2	3	2	3	BioEE 401	
	Total	26	21	15	0	
THIRD YEAR						
Midterm						
Course Code	Course Title	Unit/s	No. of Hours		Pre-requisite/s	Co-requisite/s
			Lec	Lab		
BioE 306	Muskoskeletal Mechanics	3	3	0	BioE 302	
ECE 311	Microprocessor and Microcontroller Systems and Design	4	3	3	BioE 206	
BioE 307	Programming and Modelling for BioE	1	0	3	BioE 205	
	Total	8	6	6		
FOURTH YEAR						
First Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
BioE 401	**Robotics for Biomedical Applications *****	**4**	**3**	**3**	**BioE 305**	

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BioE 402	Biomedical Laws and Professional Ethics	3	3	0	4th year standing	
BioE 404	BioE Capstone Design 2	1	0	3	BioE 403	
ENGG 105	Technopreneurship	3	3	0	4th year standing	
GEdE 101	Entrepreneurial Mind	3	3	0		
BioEE 403	BioE Elective 3	3	2	3	BioEE 402	
GEd 103	Life and Works of Rizal	3	3	0		
	Total	20	17	9		

FOURTH YEAR						
Second Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
ENGG 127	Industry Placement 2	8	800 hours			
GRAND TOTAL		181	143	66		

INDUSTRY-BASED LEARNING

INDUSTRY -BASED CURRICULUM FOR BS BIOMEDICAL ENGINEERING			
YEAR	**Semester 1**	**Semester 2**	**Midterm**
1	Academic	Academic	Academic
2	Academic	Academic	
3	Industry Placement	Academic	Academic
4	Academic	Industry Placement	**GRADUATE**